

Table 6.2: Development Tools

| <b>Tool</b>        | <b>Version</b> | <b>Purpose</b>                                    |
|--------------------|----------------|---------------------------------------------------|
| Visual Studio Code | 1.85+          | Primary code editor with Type-Script IntelliSense |
| Node.js            | 18.x LTS       | JavaScript runtime environment                    |
| npm                | 9.x            | Package manager for dependencies                  |
| Git                | 2.40+          | Distributed version control                       |
| GitHub             | –              | Repository hosting and collaboration              |
| Firebase CLI       | 13.x           | Firebase project management and deployment        |
| ESLint             | 8.x            | Static code analysis and linting                  |
| Prettier           | 3.x            | Code formatting and style enforcement             |
| Chrome DevTools    | –              | Browser debugging and performance profiling       |
| Postman            | 10.x           | API testing and documentation                     |

5. Admin can mark listings as “Recycled” after processing.
6. Carbon credits are calculated based on waste type and quantity.
7. Payment is processed through Razorpay upon delivery confirmation.
8. Notifications are sent to relevant parties at each status change.